AI Sandbox Report - Benchmark: Passengers mini

Experiment Report: Benchmark: Passengers mini

Experiment ID: `365fe6b4-43de-4277-bb70-9fa352958fed`
Generated: 2026-04-11 12:31:23 UTC

Executive Summary

Status: ✅ Completed **Task Type:** Regression **Models Evaluated:** 10 (10 passed, 0 failed constraints) **Duration:** 1.3 seconds

Best Model: Random Forest Regressor **Best Score:** 0.3756

Recommendation

Based on weighted evaluation across performance, robustness, latency, and cost, **Random Forest Regressor** is recommended with a global score of **0.3756**. **Runner-up:** Decision Tree Regressor (score: 0.3708)

Model Comparison

Rank	Model	Global Score	RMSE	MAE	R2	MAPE	Status
1	Random Forest Regressor	0.3756	37.1308	28.6691	0.9087	-	✅
2	Decision Tree Regressor	0.3708	39.2446	29.7727	0.8980	-	✅
3	Gradient Boosting Regressor	0.3691	39.8188	32.2626	0.8950	-	✅
4	XGBoost Regressor	0.3645	41.7420	33.2589	0.8846	-	✅
5	ElasticNet	0.3469	48.6381	35.8917	0.8434	-	✅
6	Lasso Regression	0.3469	48.6411	35.8970	0.8434	-	✅
7	Ridge Regression	0.3469	48.6445	35.9030	0.8433	-	✅
8	K-Nearest Neighbors Regressor	0.3388	51.3782	40.5182	0.8252	-	✅
9	LightGBM Regressor	0.3062	61.7467	49.7119	0.7476	-	✅
10	SVM Regressor	0.1350	100.8761	79.9638	0.3263	-	✅

Model Details

Random Forest Regressor

Rank: #1 | **Global Score:** 0.3756

All Metrics:

Metric	Value
mae	28.6691
mse	1378.6957
r2	0.9087
rmse	37.1308
test_mae	17.7205
test_mse	481.0134
test_r2	0.9328
test_rmse	21.9320

Decision Tree Regressor

Rank: #2 | **Global Score:** 0.3708

All Metrics:

Metric	Value

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Model Details

Random Forest Regressor

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Metric	Value
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mse	1378.6957
r2	0.9087
rmse	37.1308
test_mae	17.7205
test_mse	481.0134
test_r2	0.9328
test_rmse	21.9320

Decision Tree Regressor

Rank: #2 | **Global Score:** 0.3708

All Metrics:

Metric	Value
mae	29.7727
mse	1540.1364

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